

Rajat Kumar Mahaseth

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SUMMARY

Aeronautical Engineer with experience in computational modeling, composite material analysis, and MATLAB-based simulation. Skilled in MPSA modeling, ANSYS Fluent, Python, and Solid Edge with project experience in drone detection system and renewable energy optimization.

WORK EXPERIENCE

Intern at Buddha Air Pvt Ltd. (TIA, Kathmandu Nepal) July 2025- August 2025

Worked in the Workshop Division and assisted in Base Maintenance activities for ATR 42 and ATR 72 aircraft. Gained hands-on experience in Non-Destructive Testing (NDT) using Eddy-Current and Fluorescent Penetrant methods, along with exposure to wheel and brake overhauls and inspections.

Research Intern at Indian Institute of Technology Madras, Chennai feb 2026 – june 2026

Worked on modeling and analysis of uniaxial tensile behavior of Ramie fibre composites. Implemented MPSA model to predict stress-strain response and stiffness variation. Extracted and processed experimental data from literature and validated numerical results through comparison.

PROJECTS

Optimization of a Wind Turbine by Solidity Effect [Demo Link](#)

Designed and optimized a HAWT blade for low wind speeds using chord-wise flaps to increase solidity. Performed ANSYS Fluent simulations (k- SST) to analyze aerodynamic performance and improve efficiency and starting torque.

Design and Optimization of a Drone Detection System Using Computer Vision [Demo Link](#)

Developed a real-time drone detection system using YOLOv12 based CNN architecture to classify aerial objects into quadcopter, Helicopters, aircraft, and birds. Used Deepsort techniques to trace the drone path.

EDUCATION

2022- 2026	BE in Aeronautical Engineering at Acharya Institute, Bengaluru	(GPA: 9.2/10.0)
2020- 2022	Class 12th Kathmandu Model College	(GPA: 3.65/4.0)
2020	Class 10th NEB	(3.85/4.0)

SKILLS

Programming Tools	MATLAB, Python,
Design Software	Solid Edge
Simulation Software	ANSYS Fluent, OpenFOAM, CFD Analysis
Domain Expertise	Composite Materials, MPSA Modeling, Stress-Strain Analysis, Data Processing and Visualization

ACHIEVEMENTS

International Conference on Materials, Propulsion and Robotics (ICMPR-2025).

Participated and presented drone detection project in YUKTI fellowship program 2025.